LLM Accounts

1 second, 128 MB

This should be an easy problem. The description might be long, but you should definitely attempt on solving it.

A company has N accounts for LLM services (i.e., chatbot services) ($1 \le N \le 1{,}000$). For $1 \le i \le N$, account i has currently a quota of X_i tokens ($1 \le X_i \le 1{,}000{,}000$). (A token is a typical unit for the amount of computation in chatbot services.) LLM services have different levels of quality. Account i subscribes to an LLM service with quality Q_i ($1 \le Q_i \le 2{,}000$). The lower the quality, the cheaper the cost per token. For simplicity, a service with quality Y costs Y baht per token. **No** two accounts subscribe to services with same quality.

You are writing a program to manage how the company uses these accounts. Your program receives many requests. Each request would specifies

- the number T of tokens to be used ($1 \le T \le 1{,}000$), and
- the minimum quality R the request needs ($R \le 2{,}000$).

You want to find the most appropriate account for this request. The best account j should have enough token left, should have good enough quality (i.e., $Q_j \ge R$), and should cost the minimum amount of money, i.e., Q_j should be as small as possible.

If you can find the best account, say account j, its available token is decreased by T, and the total cost for the company is increased by T times Q_j. If there is no accounts meeting the criteria, you can discard this request (but in the end, you should report the number of requests that you have discarded as well.)

Your program will receive M requests ($1 \le M \le 1{,}000$). After you finish processing the requests, you should output the amount of money the company has to pay and the number of requests that you have discarded.

Consider the following example with N=5 and M=6. The following table describes the accounts.

i	1	2	3	4	5
X_i	100	200	200	300	100
Q_i	20	30	15	100	10

The next table describes M requests.

Request:	1	2	3	4	5	6
T	50	100	50	250	250	70
R	20	17	10	10	20	16

The M requests are processed as follows.

- Request 1: Accounts with quality >= 20 are accounts 1, 2, and 4. Account 1 has lower quality with enough tokens, so we pick account 1. After this request account 1 has 50 tokens left. You pay 50*20 = 1000 baht.
- Request 2: Accounts with quality >= 17 are accounts 1, 2, and 4. However, account 1 does not have enough tokens, so we pick account 2. After this request account 2 has 100 tokens left. You pay 100*30 = 3000 baht.
- Request 3: All accounts have quality >= 10. Account 5 (cheapest one) has enough token, so we pick account 5 and pay 50*10 = 500 baht. After this request account 5 has 50 token left.
- Request 4. All accounts have quality >= 10, but only Account 4 has enough tokens. So we pay 250*100=25000 baht. After this account 4 has 50 token left.
- Request 5: No accounts with quality >= 20 have enough token, we discard this request.
- Request 6: Account 2 is the cheapest on with quality >= 16 with enough token. We pay 70*30 = 2100 baht. After this account 2 has 30 tokens left.

After all requests are processed, we pay in total 1000+3000+500+25000+2100 = 31600 baht. We discard 1 request.

Input

The first line of the input contains two integers N and M ($1 \le N \le 1{,}000$; $1 \le M \le 1{,}000$).

The next N lines describe the accounts. For each $1 \le i \le N$, line $1+i$ contains two integers X_i and Q_i ($1 \le X_i \le 1{,}000{,}000$; $1 \le Q_i \le 2{,}000$).

The last M lines describe the requests. Each line contains two integers T and R ($1 \le T \le 1{,}000$; $1 \le R \le 2{,}000$).

Output

Your program should output two integers: the total amount that the company has to pay and the number of discarded requests.

Example

Input	Output
5 6 100 20 200 30 200 15 300 100 100 10 50 20 100 17 50 10 250 10 250 20 70 16	31600 1

This is an example in the task statement above.